

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

# 18-mavzu. Hough almashtirishi va shakl deskriptorlari

2026–2027 o'quv yili · 30 ta topshiriq

**Muhit.** Python 3.11/3.12, faqat CPU. Versiyalar qat'iy belgilangan (to'liq ro'yxat — requirements.txt): numpy==2.4.4, scipy==1.17.1, opencv-python-headless==4.13.0.92, scikit-image==0.26.0, scikit-learn==1.8.0. Tasvirlar: skimage.data.coins() (ichki) va cv2 bilan chizilgan sintetik shakllar — internet kerak emas.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
# har bir kodli topshiriq boshida:
import numpy as np, cv2
from scipy import ndimage as ndi
from skimage import data
from sklearn.neighbors import KNeighborsClassifier
SEED = 42; rng = np.random.default_rng(SEED)
def largest_contour(mask):
    cs = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_NONE)[0]
    return max(cs, key=cv2.contourArea)
L_PTS = np.array([[ -60, -40], [60, -40], [60, 0], [0, 0], [0, 50], [-60, 50]]) + 150 # "L" shakli, 300x300 niqobda
```

**Mavzu.** Hough transform ( $(\rho, \theta)$ ), akkumulyator, HoughLines, HoughLinesP, Hough circle transform), maydon, perimetr, circularity, extent, solidity, convex hull, convexity defects, Douglas–Peucker, momentlar, Hu moments, chain code, farq kodi, shape number, Fourier descriptors, matchShapes.

**Maqsadlar.** Talaba: (1) Hough akkumulyatorini, chain code, momentlar va Fourier descriptors ni qo'lda hisoblaydi; (2) invariantlik xossalarini isbotlaydi; (3) OpenCV/NumPy da chiziq va doiralarni topadi, tangalarni sanaydi, shakllarni Hu moments + k-NN bilan tasniflaydi; (4) natijani sonlar bilan asoslaydi va parametr tanlovini tushuntiradi.

**Baholash (100 ball).** A (1–14):  $14 \times 2 = 28$ ; B (15–18):  $4 \times 3 = 12$ ; C (19–30):  $12 \times 4 = 48$ ; hisobot sifati: 12. Oraliq qadamlarsiz javob — yarim ball. Kod `python -W error::DeprecationWarning` bilan xatosiz ishlashi shart.

## A. Qo'lda hisoblash ( $\rho = x \cos \theta + y \sin \theta$ ; chain code: 0 — sharq, soat miliga teskari, y yuqoriga)

- Hough ovozlari.** (1, 5), (3, 3), (5, 1) nuqtalar uchun  $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$  da  $\rho$  jadvalini tuzing. Qaysi katak 3 ovoz oladi? Chiziq tenglamasini yozing.
- Parametrlash.**  $x = 7$ ,  $y = 3$  va  $y = x + 2$  chiziqlarining  $(\rho, \theta)$  si. (5, 12) nuqtaning sinusoidasini  $\rho = A \cos(\theta - \varphi)$  ko'rinishida yozing.
- Akkumulyator hajmi.**  $800 \times 600$  tasvir,  $\Delta\rho = 1$ ,  $\Delta\theta = 0,5^\circ$ : chiziq akkumulyatori necha katak? Markaz har pikselda va radius  $r \in \{10, \dots, 40\}$  bo'lgan doira akkumulyatori-chi? Nisbatni toping.
- Hough circle (gradient).** Chegara nuqtasi (60, 40),  $r = 20$ . Gradient yo'nalishi  $\varphi = 90^\circ$  va  $\varphi = 45^\circ$  bo'lganda markaz uchun ovozlari qaysi nuqtalarga tushadi ( $a = x \mp r \cos \varphi$ ,  $b = y \mp r \sin \varphi$ )?
- Chain code.** Kontur uchlari: (0, 0), (1, 0), (2, 1), (3, 1), (3, 2), (2, 3), (1, 3), (0, 2), (0, 1) va yana (0, 0). Freeman kodi, farq kodi  $d_i = (c_i - c_{i-1}) \bmod 8$ , shape number, perimetr (juft kod 1, toq kod  $\sqrt{2}$ ) va maydon (shoelace).
- Koddan shaklga.** (0, 0) dan boshlangan kod 01234567 qanday shakl? Uchlarini, maydon, perimetr va circularity ni toping.  $90^\circ$  ga burilgan kod qanday?
- Oddiy belgilar.** "T" ko'pburchak: (0, 0), (8, 0), (8, 2), (5, 2), (5, 6), (3, 6), (3, 2), (0, 2). Maydon, perimetr, circularity, extent, convex hull uchlari va maydoni, solidity.
- Momentlar.**  $3 \times 3$  niqobda "T": piksellar  $(x, y) = (0, 0), (1, 0), (2, 0), (1, 1), (1, 2)$  ( $y$  pastga).  $m_{00}$ , centroid,  $\mu_{20}, \mu_{02}, \mu_{11}, \mu_{21}, \mu_{03}$  (qolgan 3-tartiblilar 0),  $\eta_{pq}$ , orientatsiya, Hu moments  $\phi_1 - \phi_4$ .
- Masshtab invariantligi.** Uzluksiz  $a \times b = 4 \times 2$  to'g'ri to'rtburchak:  $\mu_{20} = a^3 b / 12$ ,  $\mu_{02} = a b^3 / 12$ ,  $\mu_{00} = ab$ .  $\eta_{20}, \eta_{02}, \phi_1, \phi_2$  va eksentrisitetni toping; 3 marta kattalashtirilganda nima o'zgaradi?
- Fourier descriptors.**  $z = (0, 1, 1 + i, i)$  (birlik kvadrat) va  $z = (0, 2, 2 + i, i)$  ( $2 \times 1$  to'rtburchak) uchun  $a_k = \frac{1}{N} \sum z_n e^{-2\pi i k n / N}$  va  $d_k = |a_k| / |a_1|$  ( $k = 2, 3$ ) ni toping.
- FD invariantligi.**  $a_1 = 3 + 4i$ . Kontur  $90^\circ$  ga burilib, 2 marta kattalashtirilib, (5, -2) ga siljitsa  $a_1$  va  $|a_1|$ ,  $a_0$  qanday o'zgaradi? Boshlang'ich nuqta  $n_0$  ga surilsa-chi?
- Douglas–Peucker.**  $P_0(0, 0), P_1(1; 0, 5), P_2(2; -0, 3), P_3(3, 2), P_4(4; 1, 6), P_5(5, 0)$ ,  $\varepsilon = 0, 8$ . Har bir rekursiya qadamidagi masofalarni yozing va natijaviy sinq chiziqni toping.

- 13. Convex hull.** Nuqtalar:  $(0, 0), (4, 0), (5, 2), (3, 1), (4, 4), (1, 3), (2, 2), (0, 2)$ . Qavariq qobiq uchlarini (soat miliga teskari) va maydonini toping. Qaysi nuqtalar ichkarida?
- 14. Orientatsiya.**  $\mu_{20} = 50, \mu_{02} = 18, \mu_{11} = 12$ . Bosh o'q burchagi  $\theta = \frac{1}{2} \text{atan2}(2\mu_{11}, \mu_{20} - \mu_{02})$ , kovariatsiya xos qiymatlari (nisbiy) va eksstentritet  $e = \sqrt{1 - \lambda_2/\lambda_1}$ .

## B. Isbot va konseptual savollar

- 15. Nuqta–sinusoida ikkiligi.** Bir chiziqdagi nuqtalar sinusoidalari bitta  $(\rho^*, \theta^*)$  da kesishishini va ikki turli nuqta sinusoidalari  $[0, \pi)$  da aynan bir marta kesishishini isbotlang.
- 16. Moment invariantlari.**  $\mu_{pq}$  siljishga,  $\eta_{pq}$  masshtabga invariant ekanini;  $\phi_1, \phi_2$  burilishga invariant ekanini ( $M' = RMR^T$ , iz va determinant orqali) isbotlang. Nega  $\phi_7$  ko'zgu tasvirda ishorasini o'zgartiradi?
- 17. Fourier descriptors.** Siljish, masshtab, burilish va boshlang'ich nuqta  $a_k$  ga qanday ta'sir qilishini isbotlang;  $d_k = |a_k|/|a_1|$  nega to'rtalasiga invariant?
- 18. Konseptual.** (a) Izoperimetrik tengsizlik va nega piksel konturida uchburchak circularity si nazariydan kichik chiqadi? (b) Nega chain code faqat  $90^\circ k$  burilishga invariant? (c) Nega Hough 3 dan ko'p parametrdan qimmat va RANSAC qachon afzal?

## C. Kodli topshiriqlar (har biri alohida funksiya; ko'rsatilgan formatda chop eting)

- 19. O'z Hough akkumulyatorimiz.**  $200 \times 200$  nol tasvirga `cv2.line` bilan  $(10, 150) - (190, 150)$  va  $(20, 20) - (180, 180)$  (qalinlik 1) chizing, `E[:, 60:100]=0. hough_lines(E, n_theta=180, top=2, nms=5)->(A, peaks)` ni NumPy da yozing; bog'langan komponentlar soni, akkumulyator shakli, cho'qqilar  $(\rho, \theta, \text{ovoz})$  va `cv2.HoughLines(E, 1, np.pi/180, 80)` natijasini chop eting.
- 20. Deskew.**  $400 \times 400$  "sahifa":  $y = 60, 90, \dots, 330$  da  $(40, y) - (360, y)$  qalinligi 3 chiziqlar, `cv2.warpAffine` bilan  $a \in \{-5; 3; 11, 5\}^\circ$  ga burilgan. `estimate_skew(img): Canny(50, 150), HoughLines(edges, 1, np.pi/360, 150),  $\theta - 90^\circ$  ning medianasi.` Chiziqlar soni, topilgan qiyalik va tuzatilgandan keyingi qiyalikni chop eting.
- 21. HoughLinesP.**  $y = 60$  chiziqda  $x \in [70, 80]$  va  $[120, 150]$  uzilishlar,  $y = 140$  chiziqda  $[100, 104]$  uzilish ( $x: 10-190$ ). `HoughLinesP(E, 1, np.pi/180, 20, ...)` ni (`minLineLength, maxLineGap`)  $\in \{(20, 3), (20, 12), (20, 40), (120, 12)\}$  bilan ishga tushirib, saralangan kesmalarni chop eting. Natijani izohlang.
- 22. Hough circle detection.** `g=cv2.medianBlur(data.coins(), 5); cv2.HoughCircles(g, cv2.HOUGH_GRADIENT, dp=1, minDist=20, param1=100, param2=p2, minRadius=15, maxRadius=35), p2  $\in \{20, 25, 30, 35\}$ :` doiralarni soni va radius `min/o'rta/max`.
- 23. Tangalarni kontur bilan sanash.** `coin_mask: median 5  $\rightarrow$  Canny(50, 150)  $\rightarrow$   $3 \times 3$  closing  $\rightarrow$  ndi.binary_fill_holes  $\rightarrow$   $5 \times 5$  opening. Konturlardan  $S \geq 500$  va circularity  $\geq 0,7$  bo'lganlarini sanang; maydon min/median/max, ekvivalent radius, circularity va median bo'yicha katta/kichik tangalar sonini chop eting.`
- 24. Shakl belgilari.**  $200 \times 200$  niqobda (to'ldirilgan): doira `circle((100, 100), 60)`, kvadrat `rectangle((45, 45), (155, 155))`, to'rtburchak `rectangle((20, 70), (180, 130))`, uchburchak `(30, 150), (170, 150), (100, 29)`, 5 uchli yulduz (markaz  $(100, 100)$ , uchlari  $(100 + \rho_j \cos \alpha_j, 100 - \rho_j \sin \alpha_j)$ ,  $\alpha_j = \frac{\pi}{2} + \frac{j\pi}{5}$ ,  $\rho_j = 80$  (juft  $j$ ) yoki 32), "L" (`L_PTS=50`). Har biri uchun  $S, P, \text{circularity}, \text{extent}, \text{solidity}, \text{aspect ratio}$  va `approxPolyDP(c, 0.02P, True)` uchlarni jadval qiling.
- 25. Hu moments.** `moments_np(mask)` ( $m_{00}$ , centroid,  $\mu_{pq}$ , 7 ta Hu) ni NumPy da yozing, 8-topshiriqdagi "T" da `cv2.HuMoments` bilan solishtiring. "L" uchun  $\tilde{\phi} = -\text{sign}(\phi) \log_{10} |\phi|$ ; burilish  $30^\circ, 90^\circ, (45^\circ, 0, 6)$  va ko'zgu `([:, ::-1])` da `max | $\Delta \tilde{\phi}_{1..6}$ |` va  $\phi_7$  ishorasini chop eting.
- 26. Hu moments + k-NN.** 4 sinf (doira, kvadrat, uchburchak, yulduz),  $128 \times 128$ , tasodifiy burilish, masshtab `[0, 6; 1, 3]`, siljish  $\pm 10$ , uchlarga  $\pm 15\%$  shovqin (`seed=42`); train 60, test 30 ta/sinf. `StandardScaler + KNeighborsClassifier, k  $\in \{1, 3, 5\}$` , belgilar  $\tilde{\phi}_{1..4}$  va  $\tilde{\phi}_{1..7}$ : aniqlik va confusion matrix.
- 27. Chain code.**  $12 \times 12$  niqobga `fillPoly([[2, 2], [7, 2], [9, 4], [9, 8], [5, 9], [2, 6]])`; `CHAIN_APPROX_NONE` konturidan kod ( $y$  yuqoriga), farq kodi, shape number, perimetr va `arcLength`, siqish (bit). `np.rot90` dan keyin shape number tekshiruv.
- 28. Fourier descriptors.** Konturni yoy uzunligi bo'yicha  $N = 64$  nuqtaga qayta diskretlang;  $d = |a_k|/|a_1|$  ( $k = 2..9$  va  $-8..-1$ ). "L" ning burilgan/kichraytirilgan nusxalari va boshlang'ich nuqta siljishida farq;  $|k| \leq K$  ( $K = 2, 4, 8, 16, 32$ ) bilan tiklash xatosi; L, kvadrat, doira orasidagi masofalar.
- 29. Convexity defects.** Sintetik "qo'l": kaft `circle((130, 170), 55)`, 4 barmoq `line` (qalinligi 22, burchaklar  $-50, -18, 12, 42^\circ$ , uzunligi 120), bosh barmoq  $(-100^\circ, 95, \text{qalinligi } 24)$ . `Solidity`, nuqsonlar soni, chuqurligi  $> 40$  px bo'lganlar va barmoqlar soni.
- 30. matchShapes.** "L" ni uning burilgan ( $30^\circ$ ),  $0,6 \times$ , ko'zgu nusxalari va kvadrat, to'rtburchak, uchburchak, yulduz, doira bilan `CONTOURS_MATCH_I1/I2/I3` bo'yicha solishtiring; I1 bo'yicha eng yaqin 3 tasini chop eting. Ko'zgu nega 0?

## Hujjatlar

- OpenCV Hough (HoughLines, HoughLinesP, HoughCircles): [https://docs.opencv.org/4.x/dd/d1a/group\\_\\_imgproc\\_\\_feature.html](https://docs.opencv.org/4.x/dd/d1a/group__imgproc__feature.html)
- OpenCV shape (moments, HuMoments, matchShapes, convexHull, approxPolyDP): [https://docs.opencv.org/4.x/d3/dc0/group\\_\\_imgproc\\_\\_shape.html](https://docs.opencv.org/4.x/d3/dc0/group__imgproc__shape.html)
- OpenCV Hough Line tutorial: [https://docs.opencv.org/4.x/d6/d10/tutorial\\_py\\_houghlines.html](https://docs.opencv.org/4.x/d6/d10/tutorial_py_houghlines.html)
- OpenCV Hough Circle tutorial: [https://docs.opencv.org/4.x/da/d53/tutorial\\_py\\_houghcircles.html](https://docs.opencv.org/4.x/da/d53/tutorial_py_houghcircles.html)
- OpenCV contour features: [https://docs.opencv.org/4.x/dd/d49/tutorial\\_py\\_contour\\_features.html](https://docs.opencv.org/4.x/dd/d49/tutorial_py_contour_features.html)
- scikit-learn KNeighborsClassifier: <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>